

CODE WALKTHROUGH & ARCHITECTURE

Food Assistant App

A Domain-Locked Culinary AI Application built with Streamlit, LangChain, and OpenAI GPT Models

System Overview & Core Philosophy

🎯 Single-Domain Focus

The application strictly locks the LLM to food, recipe, cooking, and culinary topics. Off-topic queries are gracefully redirected without compromising system context or security rules.

🛡️ Interactive & Secure

Incorporates an active API key validation gate before granting chat access, ensuring users provide valid OpenAI credentials while keeping runtime session memory isolated.

Strict Domain Locking Enforcement

-  **System Prompt Primacy:** Injects `FOOD_DOMAIN_SYSTEM_PROMPT` as message index 0, rendering domain overrides impossible.
-  **Prompt Injection Resistance:** User style adjustments in the sidebar are strictly appended as non-overriding constraints.
-  **Strict Guardrails:** Rejects all non-culinary topics (e.g. coding, math, general advice) instantly with a polite redirection.



Key Technical Capabilities



API Key Authentication

Live API validation via `client.models.list()`. Prevents unauthorized API consumption and invalid key runtime crashes.



Conversational Memory

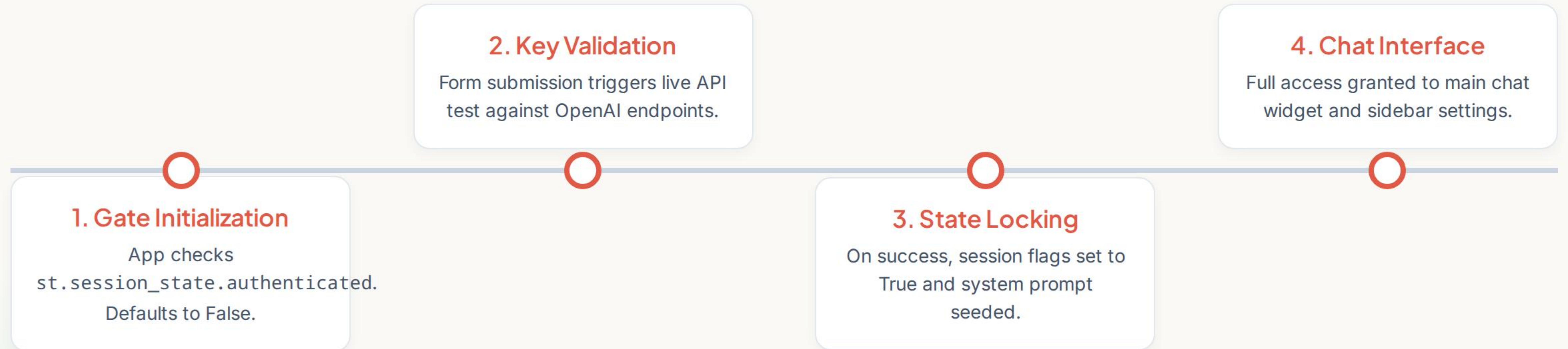
Uses LangChain's `HumanMessage`, `AIMessage`, and `SystemMessage` wrappers within `st.session_state` to maintain chat history.



Dynamic Sidebar Controls

Allows users to adjust custom persona style notes (e.g., 'casual tone', 'vegan focus') on-the-fly without breaking core rules.

Authentication & App Lifecycle



Core Modules & Dependencies

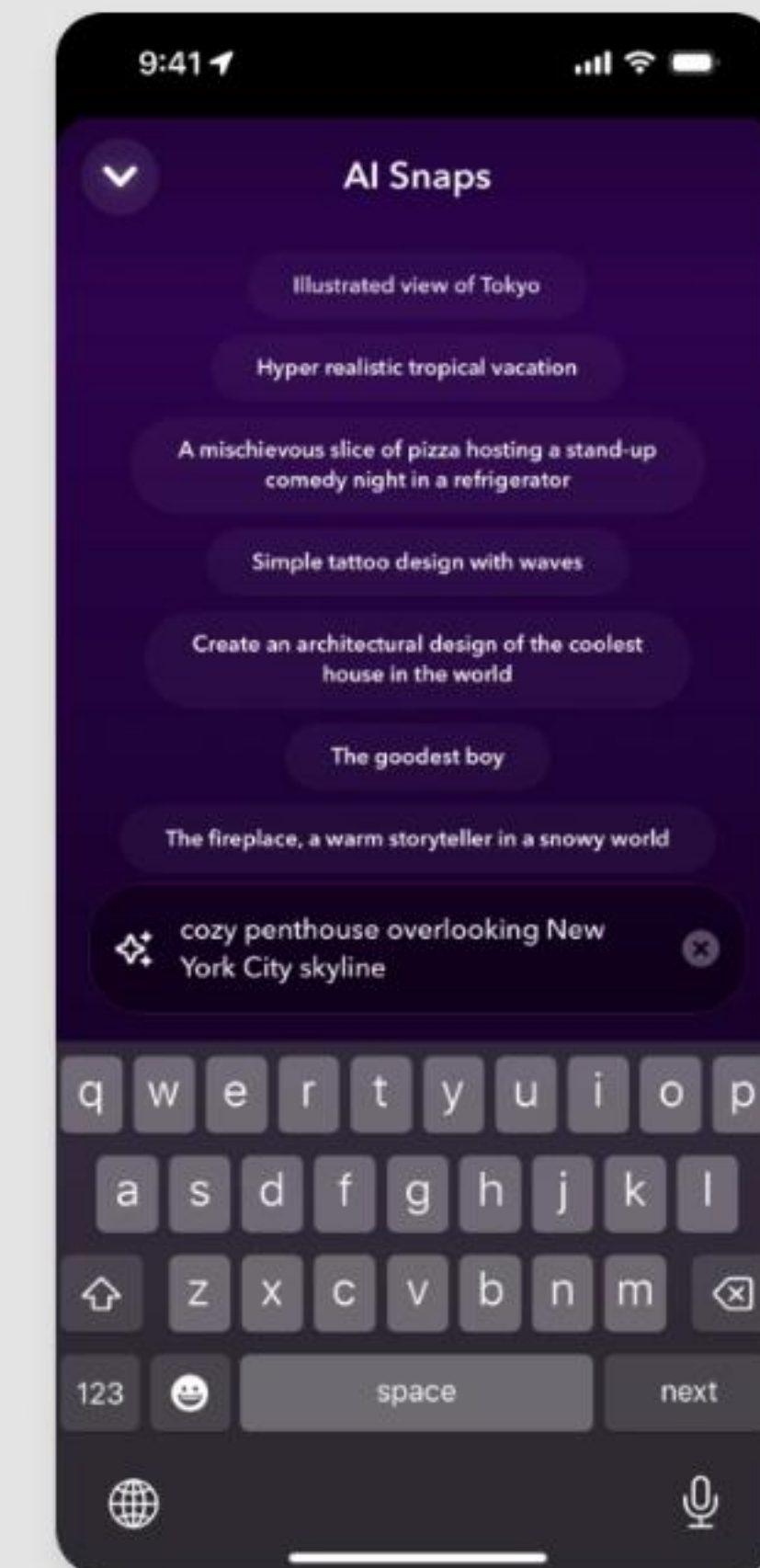
Module / Library	Role in Application	Key Functions Used
Streamlit (st)	Web interface, layout containers, session management	<code>st.session_state</code> , <code>st.chat_input</code> , <code>st.form</code>
LangChain Core	Structured prompt messaging schema	<code>HumanMessage</code> , <code>AIMessage</code> , <code>SystemMessage</code>
LangChain OpenAI	LLM model wrapper & invocation	<code>ChatOpenAI(model="gpt-3.5-turbo")</code>
OpenAI Native SDK	API Key validation before chat instantiation	<code>openai.OpenAI().models.list()</code>

Session State Architecture

-  **st.session_state.api_key:** Temporarily holds the validated key during the active browser session.
-  **st.session_state.authenticated:** Boolean toggle switching between Gate 1 (Auth) and Gate 2 (Chat UI).
-  **st.session_state.messages:** List of LangChain message objects storing entire chat thread starting with SystemMessage.
-  **st.session_state.extra_instructions:** Custom tone string dynamically reconstructed into system messages.

User Experience & Interface Design

- ⚡ **Instant Feedback:** Spinner indicators keep users informed during key validation and model generation.
- ↺ **Session Reset:** Integrated "Change API Key" sidebar button clears keys and resets chat state cleanly.
- 🍴 **Gourmet Visuals:** Clean emoji iconography and custom titles tailored specifically for culinary guidance.



System Metrics & Reliability

100%

Domain Enforced

Key Operational Highlights

Zero Unauthenticated Leaks: Gate 1 ensures no API queries can be submitted without prior validation.

Stateless Persistence: No sensitive keys or user message histories are stored on disk or in database storage, maintaining strict privacy.

Questions & Answers

Explore the `streamlit_chatgpt.py` implementation or test your food queries!

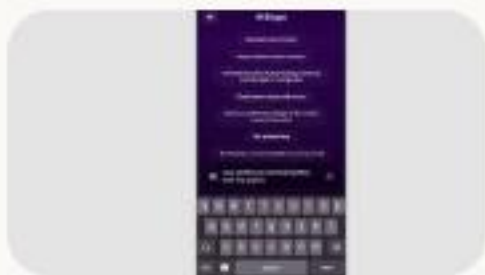
 Streamlit + LangChain Integration

Image Sources



<https://i.pcmag.com/imagery/articles/07dxfKsJoAFTIySEVXRfNQ2-1..v1756475832.jpg>

Source: www.pcmag.com



https://cdn.prod.website-files.com/5f16d69f1760cdba99c3ce6e/679a08ddc99c7a3360819b56_AD_4nXeC_9FBeteWMt_CyPVV8JdlPEUJ279N3esN6lgEJ63hZ0aDc0cl7Sn2-_U06N61OJGsJzw-Nz8rRIU2oyUSZb16KgmRYNKyc6NWthIBABS3dO0owWwRHmDBlozgZOkpXToffuKZQw.png

Source: www.eleken.co